SUMMARY

FAIZ NI'MATUL HAQ. *Patrol Boat Design for Enforcement of Illegal Fishing*. (Under the guidance of **Dr. Ir. Tri Djoko Lelono, M.Si** and **Sunardi, S.T, M.T**)

Indonesia has abundant natural resource potential because it is located in a strategic position. This very strategic position makes Indonesia a very important route for domestic and international shipping. However, this potential has not been utilized properly due to the rampant acts of theft of marine resources by foreign ships. Problems with unclear status and boundaries as well as weak supervision and security in Indonesian waters are one of the causes of illegal fishing. One of the efforts or strategies to support the defense and security of Indonesian territorial waters can be carried out by tightening guard and supervision by using patrol boats. Patrol Boats are ships that are included in fast boats by carrying out security and guard missions for ships operating in Indonesian waters. However, until now monitoring activities are still not optimal. This is because illegal fishing activities still often occur, so along with the development of technology and information, surveillance using patrol boats needs to be improved, one of which is by designing a design concept.

This research was conducted from September to November 2022. The purpose of this research was to design a patrol boat design concept and innovation that could be applied to support the pursuit of illegal fishing and conduct a technical feasibility analysis. The research method used is descriptive method using secondary data in the form of comparison ships and journals as a literature study. The feasibility analysis carried out is the stability and resistance of the ship. Stability calculations were carried out for 2 load conditions, namely in Loadcase 1 using fuel oil (BBM), fresh water tanks were 100% full and Loadcase 2 used fuel oil (BBM), fresh water tanks were 50% full. Resistance analysis was carried out using the Savitsky pre-planning and planning methods with an efficiency of 60%.

The results of the ship's feasibility analysis show that the stability analysis meets the standard values set by the IMO (International Maritime Organization) code on Intact stability A.749 (18), Ch 3 - design criteria applicable to all ships. Analysis of ship resistance using the Savitsky Planning method with an efficiency of 60% with a maximum speed variation of 40 knots requires a power of 110.105 kN and a maximum power of 7224.05 HP, so the selected engine is the Marine Diesel Engine Yuchai Motor Marino 3000 HP. The results obtained from Stability and resistance analysis as well as data from comparison vessels, patrol boats that are designed to be feasible to operate.